

CSCA67/MATA67 Week 2

Discrete Mathematics

Anna Bretscher

Today

- ✓ Review Sum and Product Rules
- ✓ Principle of Inclusion Exclusion
- ✓ Pizza!!
- ✓ Combinations
- ✓ Permutations with Repeated Items
- ✓ Combinations with Repetition $\Leftarrow$ time permitting

Practice! Practice! Practice!

Review - Sum and Product Rules

The Sum Rule (general).

If an *operation* can be performed in n different ways, each having x_i possible *outcomes* then the *total number* of possible *outcomes* is:

$$\sum_{i=1}^n x_i$$

The Product Rule (general).

If an *operation* takes k steps, and

- the *first step* can be performed in x_1 ways,
- the *second* in x_2 ways and
- ... the k^{th} in x_k ways,

then the *total number* of ways *the operation* can be performed is: $\prod_{i=1}^n x_i$

Password Example Again

Generating passwords

Suppose a *password* must be between *6 and 10* characters long and may include *letters* and *a digits*. How many *passwords* are possible?

The solution uses the *Product* rule and the *Sum* rule:

$$62^6 + 62^7 + 62^8 + 62^9 + 62^{10}$$

Q. Can you see where?

Q. When do we need to be careful with the Sum rule?

Example

How many bit strings of length eight either begin with 00 or end with 101?

00110100, 10100101, 11001011

Solution?

Q. How many begin with 00?

A. 2^6

Q. How many end with 101?

A. 2^5

Q. Why isn't $2^6 + 2^5$ the correct answer?

A. There are strings that both begin with 00 and end with 101 so are counted twice...

Therefore, the correct answer is:

$$2^6 + 2^5 - 2^3 = 88$$

How did we get this?

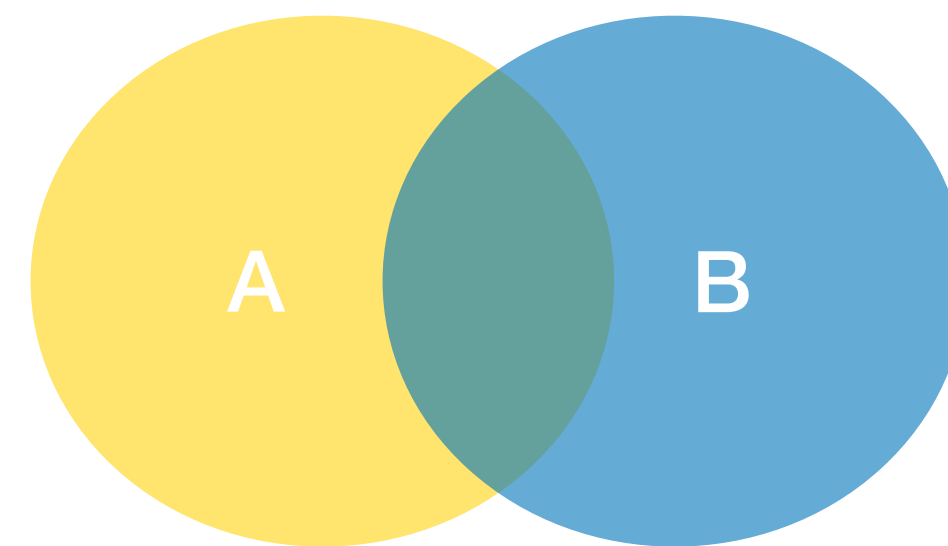
Inclusion - Exclusion Principle of Sets

Inclusion Exclusion Principle of Sets

Let A and B be two sets.

Let $n(X)$ represent the number of elements in the set X .

*Then the number of elements in the **union** $A \cup B$, (i.e., the elements in A or B) is:*



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Pizza!

Pizza Video (1997)

<https://www.youtube.com/watch?v=xak-onyKwyA>

Recap

- 2 pizzas
- up to 5 toppings on each
- 11 toppings to choose from
- all for \$7.98 (back in 1997 before most of you were born!!).

The commercial's **math kid** claimed there are **1,048,576** possibilities.

Let's do the *calculation ourselves* and verify it's **correctness**...or is it??

Making The First Pizza

There are 11 toppings.

For the pizza we can select up to 5 toppings for our pizza.

Q. How many ways can the pizza be made with 11 toppings to choose from?

Q. Does order matter?

A. No.

Q. Is there repetition?

A. No.

Q. How many 0 topping pizzas can we make?

A. 1

Making The First Pizza

There are 11 toppings.

For the pizza we can select up to 5 toppings for our pizza.

Q. How many 1 topping pizzas can we make?

A. 11

Q. How many 2 topping pizzas can we make?

A. $11 \cdot 10$?

⇒ we've over counted...pepperoni and mushrooms is the same pizza as mushrooms and pepperoni.

A. $\frac{11 \cdot 10}{2}$

Making The First Pizza

There are 11 toppings. For the pizza we can select up to 5 toppings for our pizza.

Q. How many 3 topping pizzas can we make?

A. $11 \cdot 10 \cdot 9$?

⇒ we've over counted again...pepperoni, green peppers and mushrooms is the same pizza as mushrooms, green peppers and pepperoni.

Q. How much have we over counted by?

Lets look at all the ways we can order pepperoni, green peppers and mushrooms:

PGM PMG

MPG MGP

GMP GPM

Making The First Pizza

Q. How much have we over counted by?

PGM PMG

MPG MGP

GMP GPM

This is the number of ways to order 3 items...so...

3!

Q. How many 3 topping pizzas can we make?

A. $\frac{11 \cdot 10 \cdot 9}{3!}$

Q. How can we rewrite this using only factorials?

A. $\frac{11!}{3!8!}$

Making The First Pizza

Q. How many 4 topping pizzas can we make?

A.
$$\frac{11 \cdot 10 \cdot 9 \cdot 8}{4!}$$

Q. How can we rewrite this using only factorials?

A.
$$\frac{11!}{4!7!}$$

Q. What is the pattern?

A.
$$\frac{11!}{4!7!} = \frac{11!}{4!(11-4)!} \Rightarrow \frac{n!}{r!(n-r)!}$$

Combinations

Take Home

The formula to select r of n objects *without repetition* when *order doesn't matter* is:

$${}_nC_r = C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

The number of pizzas we can make using up to 5 toppings from 11 toppings is:

$$C(11,0) + C(11,1) + C(11,2) + C(11,3) + C(11,4) + C(11,5)$$

$$= \sum_{i=0}^5 C(11,i) = \sum_{i=0}^5 \frac{11!}{(11-i)!i!}$$

Back to Pizza!

Recap

- 2 pizzas
- up to 5 toppings on each
- 11 toppings to choose from
- all for \$7.98 (back in 1997 before most of you were born!!).

The commercial's *math kid* claimed there are **1,048,576** possibilities.

Let's do the *calculation ourselves* and verify it's *correctness*...or is it??

We can pick one pizza in

$$\sum_{i=0}^5 C(11,i) = 1 + 11 + \frac{11 \cdot 10}{2} + \frac{11 \cdot 10 \cdot 9}{3!} + \frac{11 \cdot 10 \cdot 9 \cdot 8}{4!} + \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7}{5!}$$

= 1024 ways

Back to Pizza!

What about two pizzas?

- 2 pizzas
- up to 5 toppings on each
- 11 toppings to choose from
- all for \$7.98 (back in 1997 before most of you were born!!).

On the board!



Setting Up NHL Divisions

Suppose there are only 6 NHL teams;

Edmonton, Calgary, Winnipeg, Toronto, Montreal and Ottawa.

And two divisions A and B, 3 will play in A and 3 will play in B.

Q. How many ways can the 6 teams be split between the two divisions?

Let M = Montreal, O = Ottawa and T = Toronto

Let's look at the case where Montreal, Ottawa and Toronto are selected to play in Division A.

Q. How many different orderings are there of the 3 teams?

A. $3! = 6$

M O T
M T O
T M O
T O M
O T M
O M T

Setting Up NHL Divisions

Suppose there are only 6 NHL teams;

Edmonton, Calgary, Winnipeg, Toronto, Montreal and Ottawa.

Let's finish the problem...

The number of ways to select teams to play in division A is:

$$6 * 5 * 4$$

This *over counts* by ?

$$3!$$

Which gives us

$$\frac{6 * 5 * 4}{3!} = \frac{6!}{3!3!} = C(6,3)$$

Q. Are we done? Do we need to count the number of ways to select teams to play in B?

Non-distinct Objects

Q. What happens if we *allow some items to be repeated*?

Example. How many *7 digit phone numbers* can be made from the *digits 1,1,2,3,4,4,4*?

Q. Does *order* matter?

A. Yes, so a *permutation*.

The formula for a *permutation* with *distinct objects* is

$$P(n, r) = n!/(n-r)!$$

We can convert this to a problem we *can* solve. Colour code the *non-unique* digits to make them *unique*.

1,1,2,3,4,4,4

Non-distinct Objects

Q. How many permutations of the unique digits 1,1,2,3,4,4,4 are there?

A. $P(n, r) = P(7, 7) = 7!/1! = 7!$

Let's try to solve the original problem.

Which permutations of 1,1,2,3,4,4,4 have the same digit ordering as 1,1,2,3,4,4,4?

Q. How many ways can we reorder the 1s?

A. 2.

Q. How many ways can we reorder the 2 or the 3?

A. 1

Q. How many ways can we reorder the 4s?

*A. $3*2*1 = 3! = 6$*

Non-distinct Objects

Let's look at the *permutations* of $1, 1, 2, 3, 4, 4, 4$ that have the same ordering of 1's, 2's, 3's and 4's:

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

$1, 1, 2, 3, 4, 4, 4$

Observation.

The number of permutations of $1, 1, 2, 3, 4, 4, 4$ is larger than the number of permutations of $1, 1, 2, 3, 4, 4, 4$.

Q. How much larger?

Non-distinct Objects

Notice that the number of permutations of 1,1,2,3,4,4,4 should equal:

The unique permutations of 1,1,2,3,4,4,4 $\times$ the number of ways to arrange the 1s, 4s, 2s and 3s.

Q. How many ways can we arrange the 1s, 4s, 2s and 3s?

A. 2!, 3!, 1!, 1!

Q. How many ways can we arrange 1,1,2,3,4,4,4?

A. 7!

So,

$$(\text{unique permutations of } 1,1,2,3,4,4,4) \times 2! \times 3! \times 1! \times 1! = 7!$$

$$(\text{unique permutations of } 1,1,2,3,4,4,4) = \frac{7!}{3!2!1!1!}$$

Non-distinct Objects

*Q. How do we say this **mathematically**?*

*A. We represent this **formula**, the unique **permutations** of 1,1,2,3,4,4,4 by*

$$P(7; 3, 2, 1, 1) = P(7; 3, 2)$$

When a **set of objects** has **repeated** items that we permute we list the **quantities of each** of the items in our permutation notation in **decreasing** order.

For brevity, we **drop** the **1s** from the **formula**.

Review

Permutations/Arrangements with repetition

$$n^r$$

Permutations/Arrangements without repetition

$$P(n, r) = \frac{n!}{(n - r)!}$$

Permutations/Arrangements with repeated items

$$P(n; r_1, r_2, \dots, r_m) = \frac{n!}{r_1! r_2! \cdots r_m!}$$

Combinations without repetition

$$C(n, r) = \frac{n!}{(n - r)! r!}$$

Combinations with Repetition

Donut Example

We are buying 12 donuts and have 4 flavours to choose from.

Chocolate	Sprinkle	Old Fashioned	Vanilla
XXX	XX	XXXXXX	XX
	XXXXXX	XXXXXXXX	



We can restate this problem as an arrangement with repetition of Xs and |s.

Q. How many Xs are there? How many |s are there?

A. Xs = # donuts = 12 and # flavours - 1 = 3 = |s.

Combinations with Repetition

Donut Example

Q. How many ways can we arrange 12 Xs and 3 Is?

A. This is *permutation with repeated items*... $P(T; r_1, r_2)$

Q. What is our total number of items T ?

A. $12 + 3 = 15$

Q. What are the values of r_1 and r_2 ?

$$P(T; r_1, r_2) = P(12 + 3; 12, 3) = P(15; 12, 3) = \frac{15!}{12!3!}$$

Q. How is this related to *combinations*?

Combinations with Repetition

Q. How is this related to *combinations*?

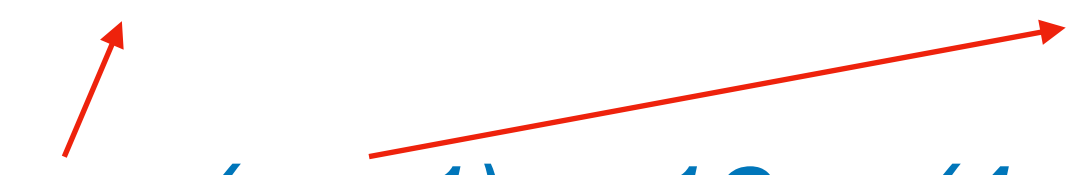
$$P(12 + 3; 12, 3) = P(15; 12, 3) = \frac{15!}{12!3!} = C(15, 12)$$

A. Since $T = r_1 + r_2$,

$$P(T; r_1, r_2) = \frac{T!}{r_1!r_2!} = \frac{T!}{r_1!(T - r_1)!} = C(T, r_1)$$

Q. Notice that r_2 is the *# types - 1* = *# flavours - 1* = $4 - 1 = 3$ and

T is the *# of repeated selections + # of types - 1*


$$T = r + (n - 1) = 12 + (4 - 1)$$

$$C(T, r_1) = C(r + (n - 1), r) = C(r + (n - 1), n - 1)$$

Combinations with Repetition

Theorem. Given r objects and n types of objects to choose from, the number of selections with repetitions is:

$$\binom{r + (n - 1)}{r}$$

r and n can be confusing. 🤪sometimes better to think of:

$$C(\text{objects} + (\text{types} - 1), \text{objects})$$

$$= \binom{\text{objects} + (\text{types} - 1)}{\text{objects}}$$

$$= \binom{\text{objects} + (\text{types} - 1)}{\text{types} - 1}$$

Non-distinct Objects A little Deeper

Observation.

- If there are m *unique* items, we can denote the number of repetitions of the i^{th} object by r_i .
- If the total number of items is $n = r_1 + r_2 + \cdots + r_{m-1} + r_m$, then the number of ways to *arrange* the *items* is:

$n!$ = *number of ways to arrange objects with repetitions*
× *number of ways to arrange each repeated type*

$$n! = P(n; r_1, r_2, \dots, r_m) \times r_1! r_2! \cdots r_m!$$

Rearranging:

$$P(n; r_1, r_2, \dots, r_m) = \frac{n!}{r_1! r_2! \cdots r_m!}$$